

1. Parabolic Geometry & Core Equation Derivation

Geometric Setup

Let the vertex of the parabolic reflector be at the origin $(0, 0)$ and its focal point lie on the y -axis at $(0, f)$. The directrix is defined by the line $y = -f$.

By the geometric definition of a parabola, the distance from any point (x, y) on the curve to the focus equals its perpendicular distance to the directrix:

$$\sqrt{x^2 + (y - f)^2} = y + f$$

Squaring both sides:

$$x^2 + y^2 - 2yf + f^2 = y^2 + 2yf + f^2$$

$$x^2 = 4yf \implies y = \frac{x^2}{4f}$$

Rim Depth (d) and Focal Length (f) Relation

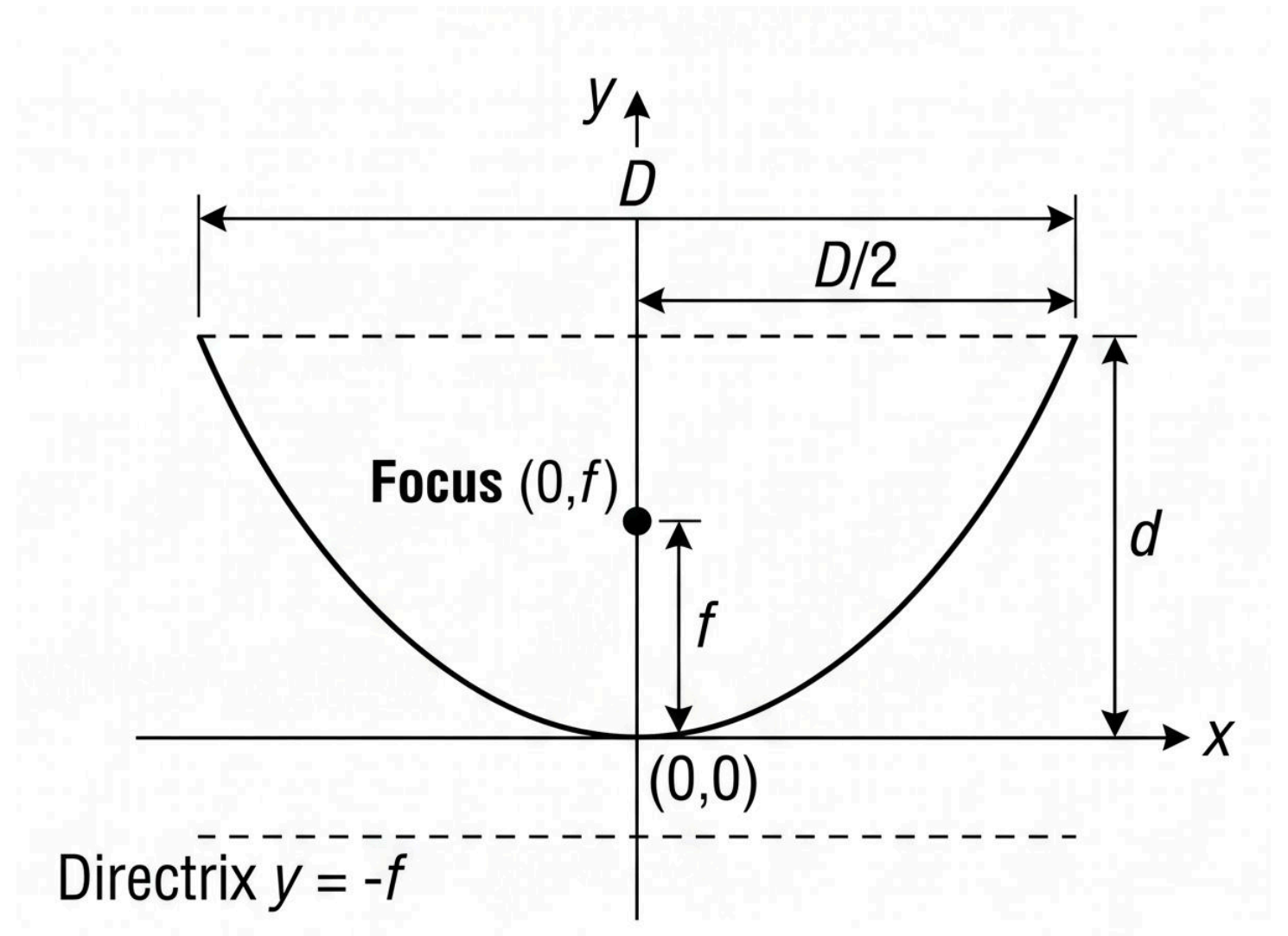
At the rim of the parabolic dish, the radius is half the total diameter ($x = D/2$) and the height is equal to the depth ($y = d$).

Substituting these boundary conditions gives:

$$d = \frac{(D/2)^2}{4f} = \frac{D^2}{16f}$$

Rearranging to solve for focal length f :

$$f = \frac{D^2}{16d}$$



2. Derivation of the Subtended Rim Angle (θ_0)

Let z_0 be the distance along the axis of symmetry from the focal point to the plane of the reflector rim. From the geometry:

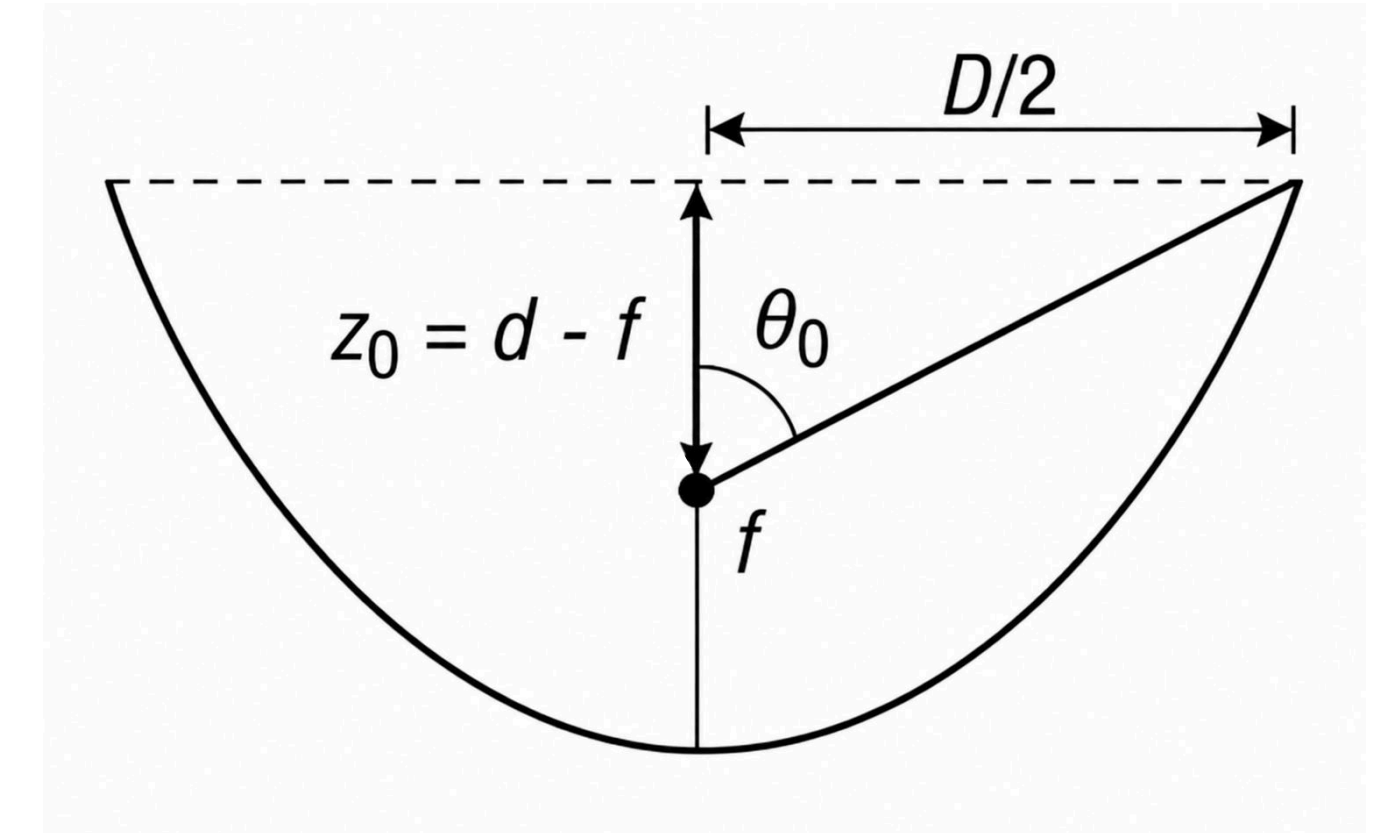
$$z_0 = d - f = \frac{D^2}{16f} - f$$

The half-angle θ_0 subtended by the reflector rim at the focus is determined using the right triangle formed by the focus, the axis, and the rim edge:

$$\theta_0 = \tan^{-1} \left(\frac{D/2}{z_0} \right) = \tan^{-1} \left(\frac{D/2}{\frac{D^2}{16f} - f} \right)$$

Dividing the numerator and denominator by D and rearranging into standard f/D dimensionless form yields:

$$\theta_0 = \tan^{-1} \left| \frac{\frac{1}{2} \left(\frac{f}{D} \right)}{\left(\frac{f}{D} \right)^2 - \frac{1}{16}} \right| \implies \theta_0 = 2 \tan^{-1} \left(\frac{1}{4(f/D)} \right)$$



Key Boundary Cases for Geometry:

Case 1 ($f/D = 0.25$)

Implies $\theta_0 = 90^\circ$ and $f = d$.

The focus lies exactly in the plane of the rim.

Case 2 ($f/D > 0.25$)

Implies $\theta_0 < 90^\circ$ and $f > d$.

The focus lies entirely outside the rim plane (shallow dish).

Case 3 ($f/D < 0.25$)

Implies $\theta_0 > 90^\circ$ and $f < D/4$.

The focus lies inside the dish cavity (deep dish).

3. Acoustic Power Interception & Absorption Model

When a plane sound wave hits the parabolic collector, the physical catchment area is its flat cross-sectional projection perpendicular to the incoming wave direction.

Projected Aperture Area

$$A_{\text{aperture}} = \frac{\pi D^2}{4}$$

Intercepted Power (P_{in})

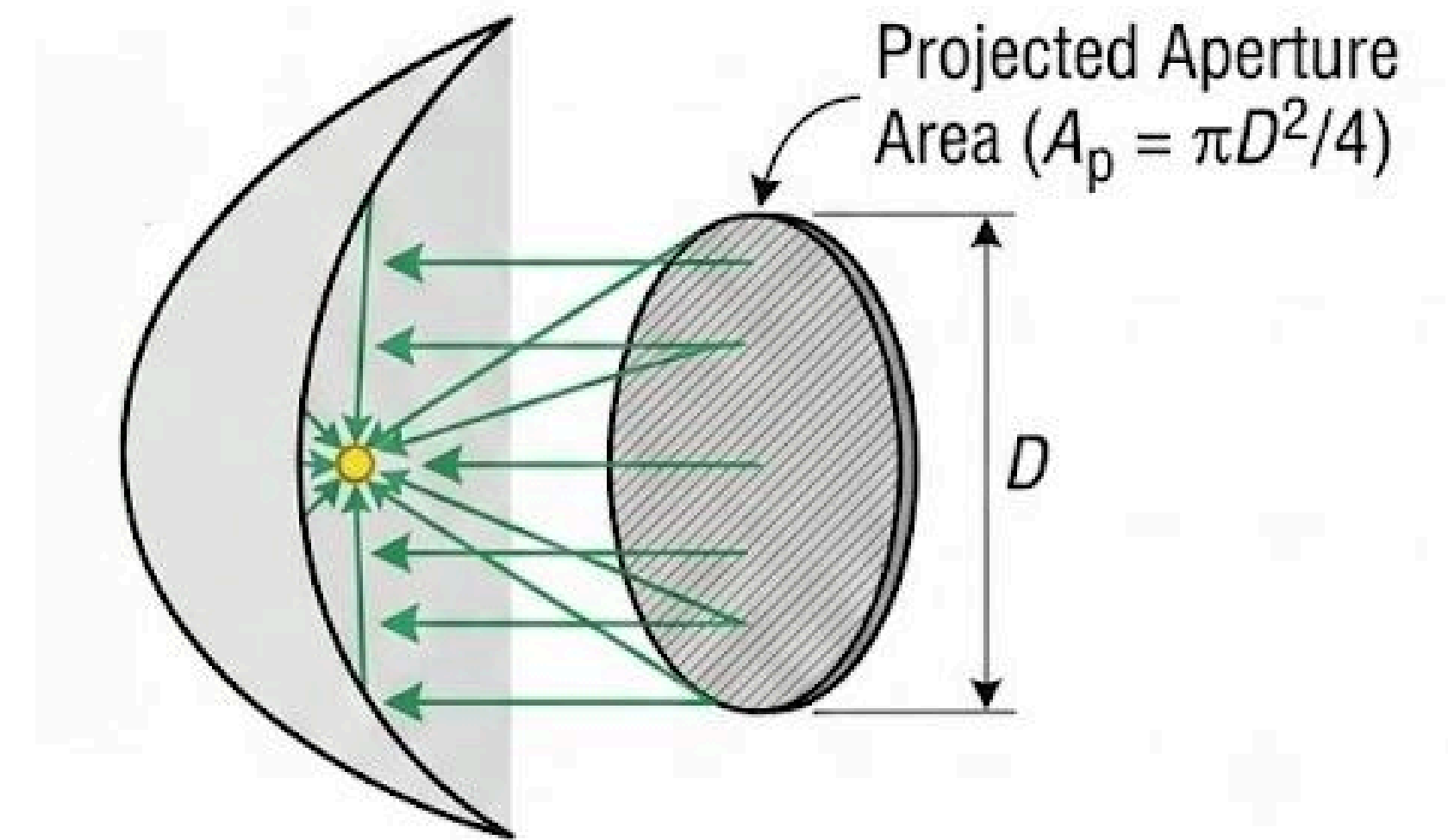
Given an incident time-averaged acoustic energy flux (intensity) I (W/m^2):

$$P_{\text{in}} = I \cdot A_{\text{aperture}} = I \cdot \frac{\pi D^2}{4}$$

Absorbed Power (P_{abs})

Accounting for the total aperture efficiency factor (η) of the system transducer setup at the focus:

$$P_{\text{abs}} = \eta \cdot P_{\text{in}} = \eta \cdot I \cdot \frac{\pi D^2}{4} \implies P_{\text{abs}} \propto D^2$$



$$P_{\text{abs}} = \eta \cdot I \cdot (\pi D^2/4) \propto D^2$$

4. Final System Optimization Constraints

To maximize the collected power while meeting the strict physical packaging restrictions of your housing unit:

Maximize Power Extraction

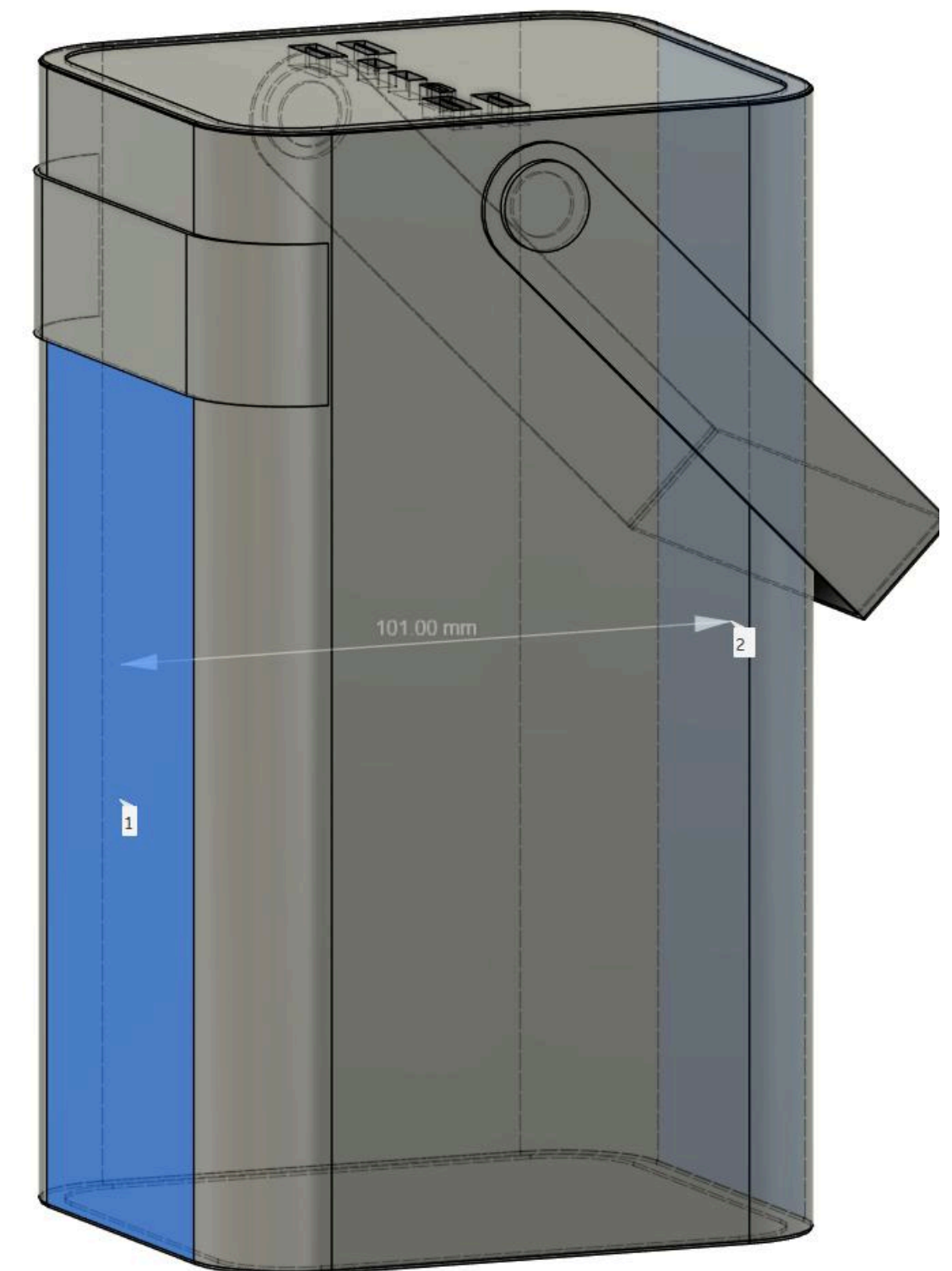
$P_{\text{abs}} \propto D^2$ (Requires maximizing D).

Volumetric Geometry Constraint

$\frac{f}{D} < 0.25$ (Ensures a deep-dish structure where the focus is fully contained inside the mirror profile to shield/secure the internal components).

Physical Spatial Constraints

- **Focal Length Target:** Keep focal length f small to minimize the depth footprint.
- **Diameter Bounds:** $105 \text{ mm} \leq D \leq 120 \text{ mm}$ (Selected lower threshold at 105 mm to clear the internal hardware/battery width safely, which stands at 101 mm).



5. Prototype : CAD Dimensions & Analytics

Inner Diameter (D): 105.00 mm

Total Dish Depth (d): 32.80 mm

A. Focal Length (f) Verification

Substituting CAD dimensions:

$$f = \frac{D^2}{16d} = \frac{(105.00 \text{ mm})^2}{16 \times 32.80 \text{ mm}} = \frac{11025}{524.8} \approx 21.00 \text{ mm}$$

B. Internal Geometric Clearance Offset (z_0)

For the deep-dish design where the focus sits completely inside the cavity:

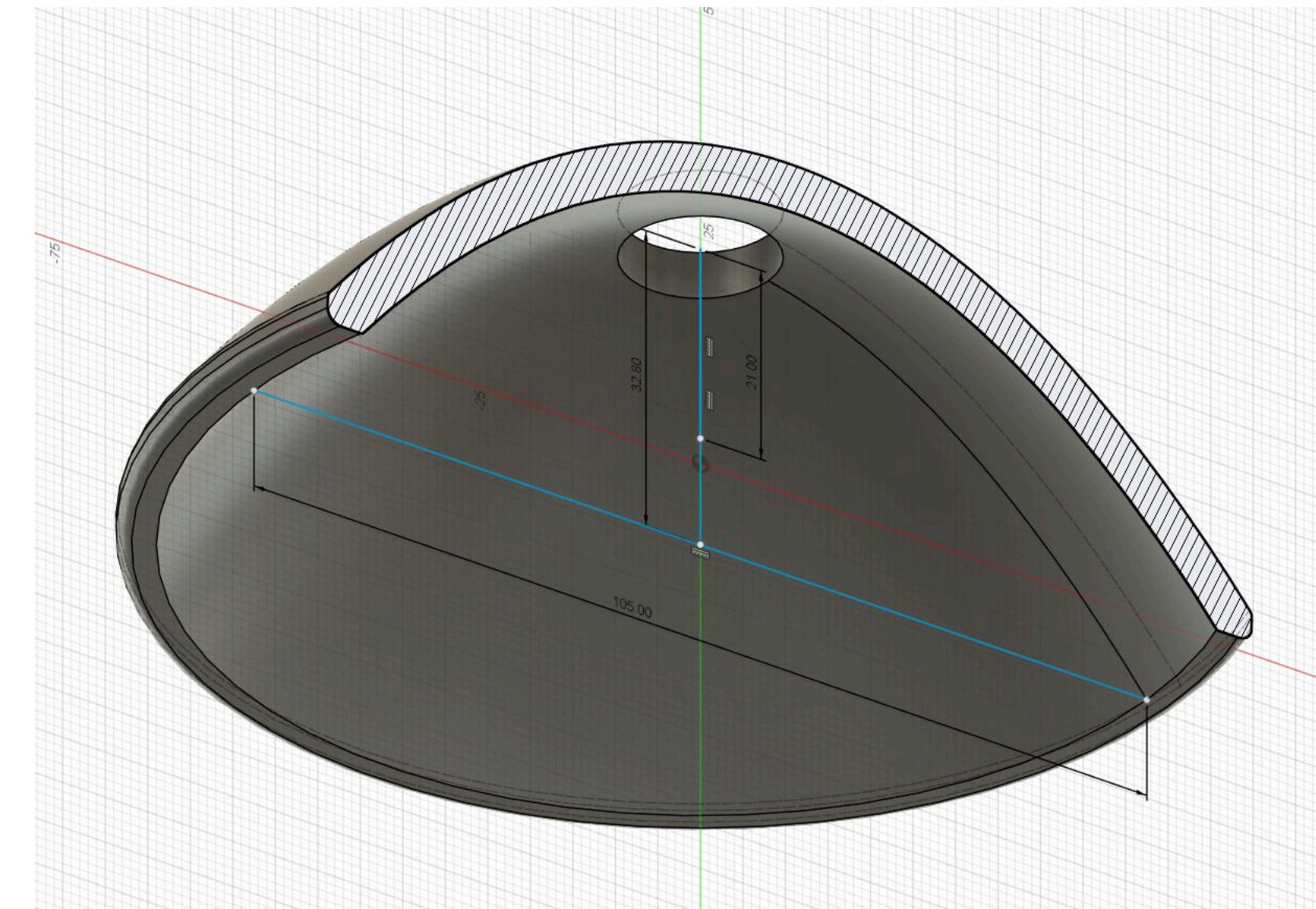
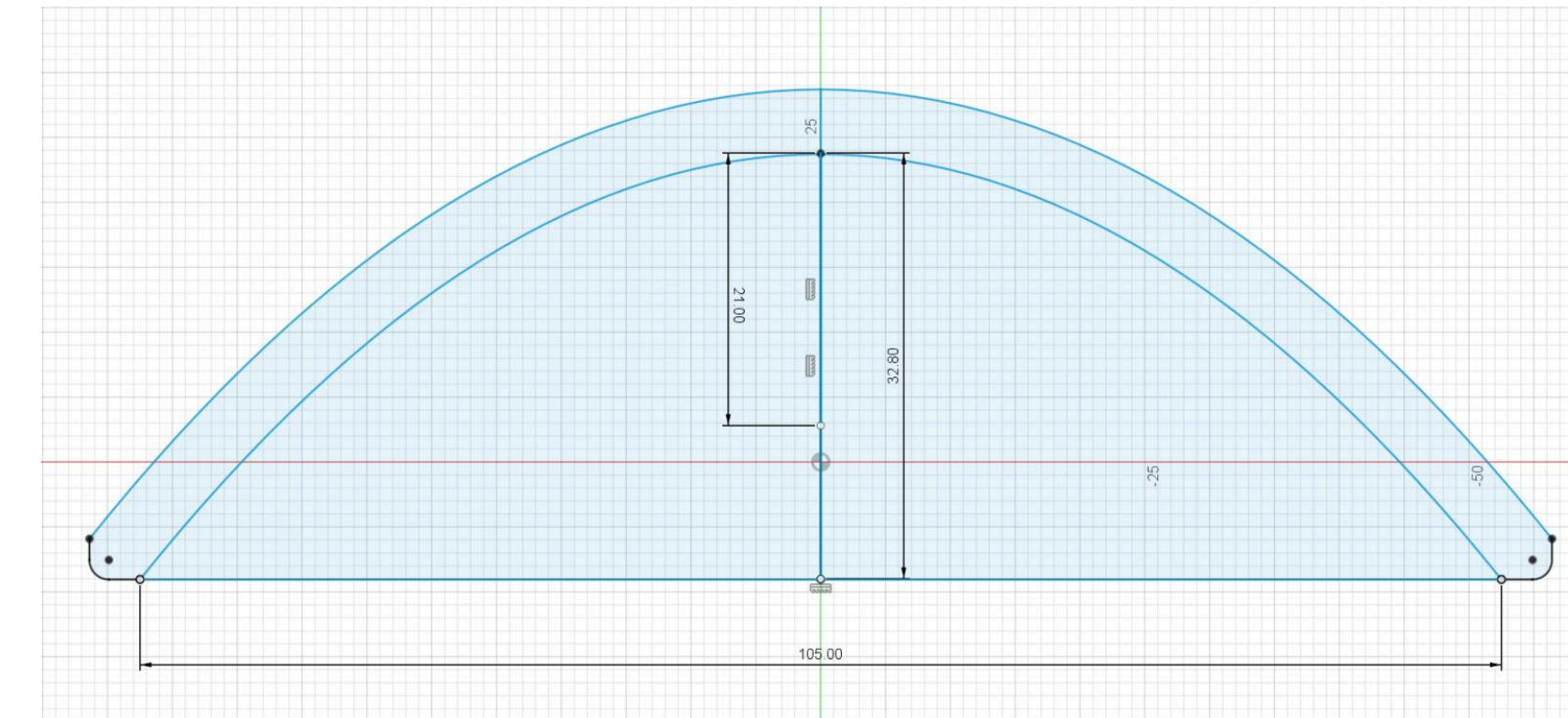
$$z_0 = d - f = 32.80 \text{ mm} - 21.00 \text{ mm} = 11.80 \text{ mm}$$

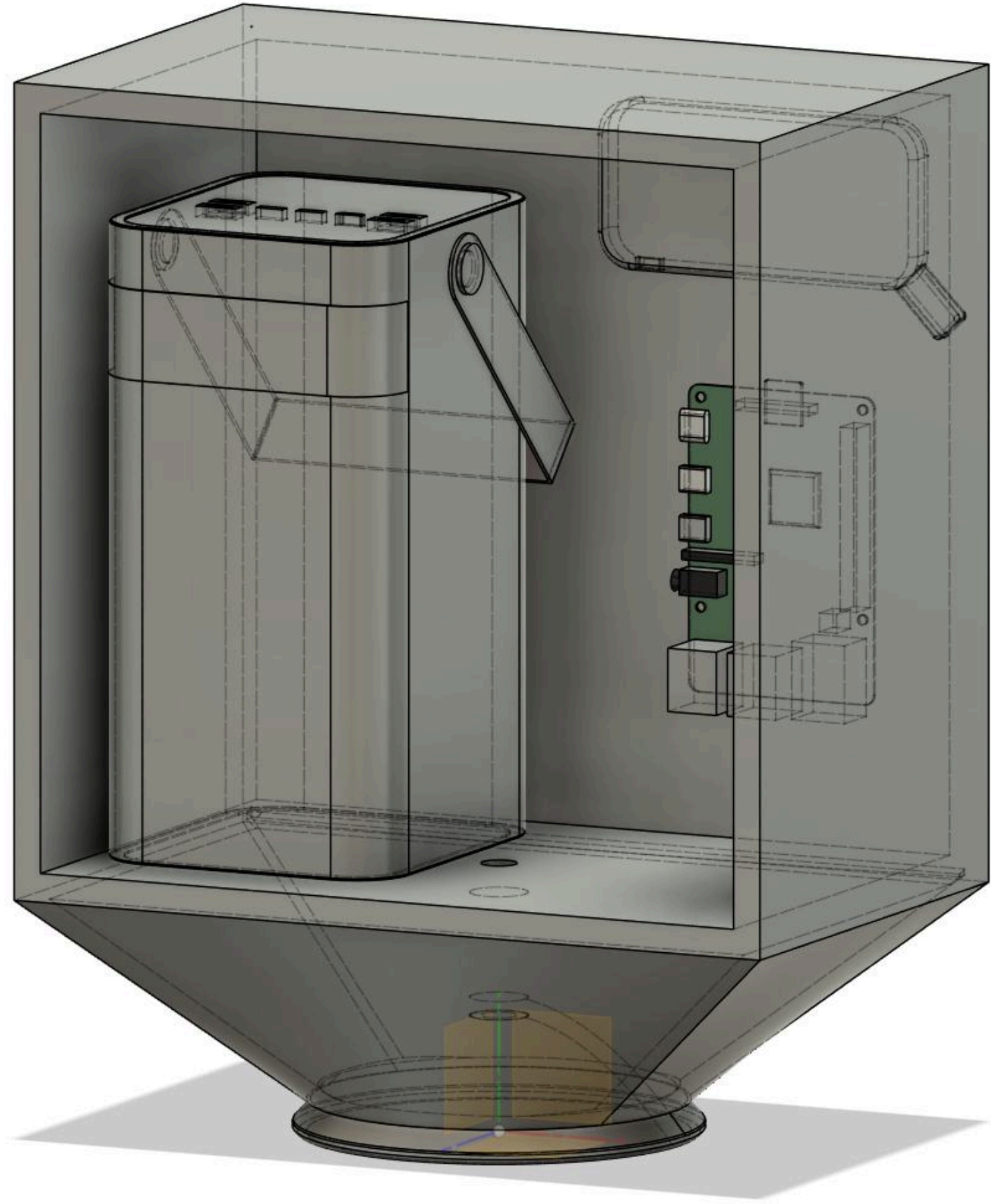
C. Dimensionless Ratio & Design Target Adherence

Calculated Ratio:

$$\frac{f}{D} = \frac{21.00 \text{ mm}}{105.00 \text{ mm}} \approx 0.20 < 0.25$$

The prototype complies with the structural constraint $f/D < 0.25$.





5. Formal Literature References

To prove this backing to your professor, you can cite these two foundational texts where these exact equations are standardized:

For the Geometry and Conic Form of Subtended Angles

Balanis, C. A. (2016). *Antenna Theory: Analysis and Design* (4th ed.). John Wiley & Sons.

Reference Note: Chapter 15, Equations (15-22) through (15-25) directly detail the transformation of the parabolic coordinate system into the $\theta_0 = 2 \tan^{-1}(1/(4(f/D)))$ relationship.

For the Acoustic Wave and Energy Aperture Formulations

Kinsler, L. E., Frey, A. R., Coppens, A. B., & Sanders, J. V. (2000). *Fundamentals of Acoustics* (4th ed.). John Wiley & Sons.

Reference Note: Chapter 5 covers acoustic intensity transformations, and Chapter 7 outlines plane wave power integration across circular barriers and collector apertures.